Position: **Any Product Role**
Case study: Data Analytics for Product
Date: December 2025

Our mission

Mo is about empowering Africans with high quality, premium refurbished phones, available for everyone. We believe that quality mobile phones help create opportunity and the way to get them in the hands of all Africans is through circular models.

Our devices are sourced and refurbished internationally and are high-quality, affordable and have a positive sustainability impact. We offer them to consumers in plans that allow for micropayment so they really are 'the smart choice'.

Case Study context

MoPhones offers smartphones to customers through installment-based credit plans and manages these accounts throughout the repayment period. Customers typically make regular payments toward their outstanding balances, and credit decisions and follow-ups play a key role in both portfolio performance and customer experience.

For this case study, you are provided with:

- Credit portfolio data
- Customer demographic and income information
- Customer satisfaction (NPS) data

The credit data is shared as point-in-time snapshots, taken at the start of January and at the end of each quarter. These snapshots reflect how customer accounts and credit outcomes look at those specific moments and are commonly used by the business to review changes in portfolio performance over time.

While MoPhones has systems in place to process payments and manage accounts, the company is still developing its analytics and reporting capabilities. As a result, there is an opportunity to use data more effectively to:

- Understand portfolio health and risk
- Track repayment and delinquency outcomes
- Assess how credit outcomes relate to customer satisfaction

This case study focuses on analysing the provided data to generate insights that can support better credit monitoring, reporting, and decision-making, while balancing portfolio performance and customer experience.

Reference: data sets

- Customer Data and Credit Data Datasets

Data Case Study

You are given credit data, customer information, and customer satisfaction (NPS) data.
The credit data is provided at specific points in time (start of January and end of each quarter).

Before starting the analysis, create the following new customer fields.

- Use the customer's Date of Birth (DOB) to calculate age as at the reporting date and group customers into these age ranges: **(18–25, 26–35, 36–45, 46-55, Above 55)**
- Add together all income-related columns , divide the total by Duration to get an average income value and group customers into these income ranges: **(Below 5,000, 5,000–9,999, 10,000–19,999, 20,000–29,999, 30,000–49,999, 50,000–99,999, 100,000–149,999, 150,000 and above)**

State any assumptions you make.

Analysis Questions

1. How does the credit portfolio perform over time, and how do payments, arrears, defaults, and account statuses vary across different customer segments?
2. Which outcomes best indicate portfolio health or risk, and what metrics would you use to track credit performance and risk over time?
3. How do credit outcomes relate to customer satisfaction (NPS), and what trade-offs, if any, exist between payment recovery and customer experience?
4. What assumptions were required due to the point-in-time nature of the data, and how do these assumptions affect your analysis?
5. What are the key limitations of the data, including missing, unclear, or inconsistent fields, and how do they impact confidence in your conclusions?
6. What improvements would you recommend to MoPhones' credit data or reporting to better monitor portfolio performance and customer experience?

Analytics & dbt Component

In addition to your analysis, you are expected to demonstrate how you would make this work repeatable and production-ready.

Using dbt, structure the provided datasets in a way that supports:

- Ongoing collections and credit analysis
- Consistent linkage between credit metrics and NPS

To demonstrate your understanding of data modeling and data lineage, your dbt project should include sample raw data inputs that represent the upstream sources you assume feed into the provided datasets.

You are free to decide:

- What raw source structures to assume and sample
- What models to create
- How to structure transformations
- What assumptions to encode
- What data quality checks are appropriate

As part of your submission, please provide:

- A link to a GitHub repository containing your dbt project, including sample raw data, models, tests, macros, and documentation.

What we want to test

- Ability to make progress without much information
- Ability to prioritize
- Analytical proficiency
- Individual contribution potential